

BFMC2026: Status Report 2

Team: SafeForce

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Github link: <https://github.com/vicyan1611/Brain/>

1. Overview of Project Status

In this phase, Team SafeForce has achieved a major milestone: **Autonomous Lane Keeping**. The vehicle can now identify lane markings and self-navigate using a vision-based approach. However, the transition from simulation/unit testing to the physical chassis revealed significant mechanical constraints and environmental challenges. We have successfully implemented a "Safety First" approach by capping motor limits to protect the hardware and are currently refining our perception algorithms to handle challenging lighting conditions on the test track.

Note on Timeline: It is important to highlight that February coincides with the Lunar New Year (Tet Holiday) in Vietnam and other Asian countries. This is the most significant holiday of the year, involving university closures and logistical pauses. Consequently, our development schedule has been adjusted to accommodate this planned break, with a sprint focus before and immediately after the holiday.

2. Current Phase

Our focus has shifted from establishing basic communication to Environmental Adaptation and Mechanical Constraints Handling.

- Perception: We are heavily investing time in fine-tuning the Computer Vision pipeline (Thresholding, Warping, Sliding Windows) to filter out noise caused by the specific material of our test map.
- Control: We are calibrating the vehicle's movement logic to operate strictly within the safe mechanical limits of the steering assembly to effectively avoid hardware damage.
- Safety: Implementing and tuning the object detection system to ensure the car stops reliably (but not prematurely) when obstacles are encountered.

3. Task Status - Overview

- Lane Keeping Algorithm (Completed): We have implemented a robust algorithm capable of detecting two lane lines simultaneously. This allows the car to calculate the center offset and navigate autonomously.
- Software Limitation for Hardware Protection (Completed): Due to mechanical interference between the steering linkage and the drive shaft, we successfully updated the embedded driver and low-level controller to clamp the operational ranges:
 - Steering Angle: Restricted to $[-15^\circ, +15^\circ]$ (Physical limit).
 - Speed: Restricted to $[-30, 30]$ (PWM/Velocity unit) to maintain stability at lower steering angles.
- Image Processing for Map Material (Completed): Since our test map (1/3 scale) is printed on decal material, it reflects ceiling lights strongly. We have optimized our image preprocessing (morphological operations and HSV thresholding) to mitigate this glare.

- Obstacle Avoidance (In Progress): The object detection logic is functional (the car stops upon detection). However, the sensor sensitivity is currently too high, causing "False Positive" stops or stopping too far from the target.

4. Blocking Points and Encountered Issues

- Mechanical Interference (Critical): We discovered a design flaw where the steering axis/linkage rubs against the drive shaft (motion axis) when the steering angle exceeds 15 degrees.
 - *Impact:* This prevents us from making sharp turns and risks damaging the servo or drive shaft.
 - *Solution:* We enforced hard software limits (Clamp function) on the steering angle. We are currently evaluating if this limited range is sufficient for the tight corners of the BFMC track or if a mechanical redesign is mandatory.
- Environmental Lighting & Map Material: The decal material used for our printed map (1/3 size) creates specular reflections (glare) from overhead lights, which the camera initially confused for white lane markings.
 - *Solution:* This required extensive effort in "tuning" the computer vision pipeline. We had to implement aggressive filtering and region-of-interest masking to distinguish real lines from light reflections.
- Sensor Sensitivity: The obstacle sensor (likely Distance/Ultrasonic or ToF) is triggering stops too early. The car currently halts well before the intended safety distance, making the behavior inefficient.

5. Next Steps

- Holiday Schedule Management: We plan to finalize the current code version before the Lunar New Year break to ensure a stable baseline when the team returns.
- Obstacle Calibration: We will implement a filter (e.g., Kalman Filter or Moving Average) for the distance sensor and adjust the stopping threshold to allow the car to approach obstacles closer before stopping, reducing false triggers.
- Cornering Strategy: Given the +/- 15-degree steering limit, we need to test if the car can navigate the sharpest turns on the official map. If not, we will implement a "differential drive" assist or a multi-point turn algorithm.
- Object Classification: Once the distance stopping is smooth, we will reintegrate the YOLO model (as prepared in our codebase) to distinguish between Stop Signs, Pedestrians, and Cars, rather than just stopping for any object.